

When the towers fall, the phone in their pocket becomes the beacon.

An instant private network that rescuers carry into a disaster zone. Any phone can send an SOS with no app, no SIM and no internet, and the survivor gets an answer back: **Delivered → Read by rescuer → Help dispatched.**

Cell towers down

Earthquakes, floods and cyclones take out power and cellular first. Survivors' phones still work; there is just no network.

One-way alerts

Cell Broadcast / SACHET can warn people but cannot hear back from someone trapped.

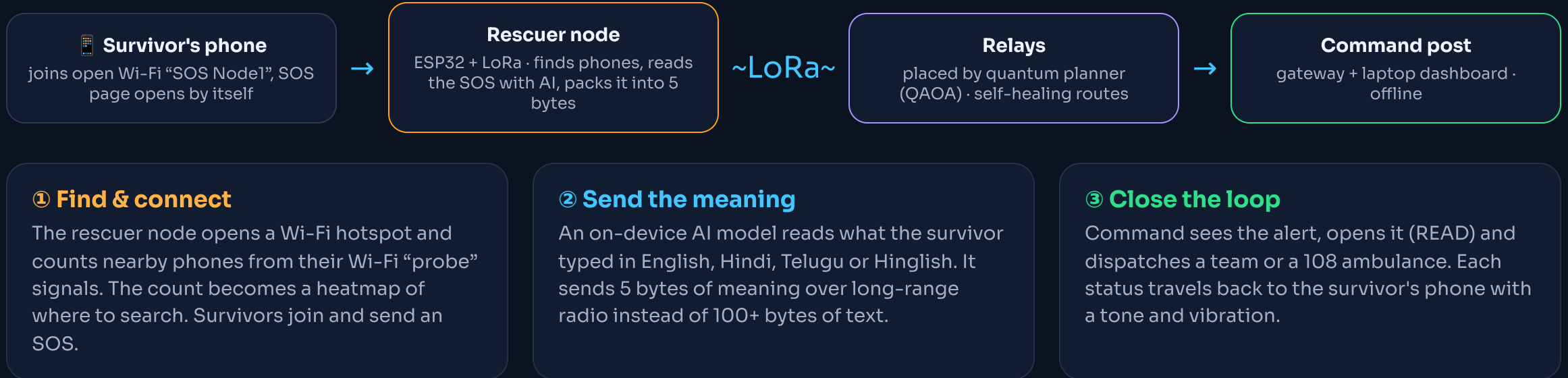
Device-per-person

Meshtastic, goTenna and rescue watches need hardware bought and set up before the disaster.

No closed loop

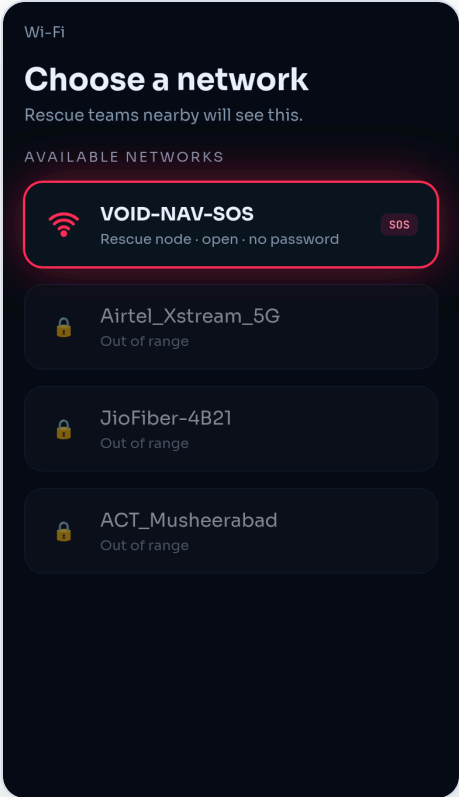
Even when an SOS gets out, the survivor never learns it was read or that help is coming.

A rescue kit that turns any phone into a two-way SOS beacon

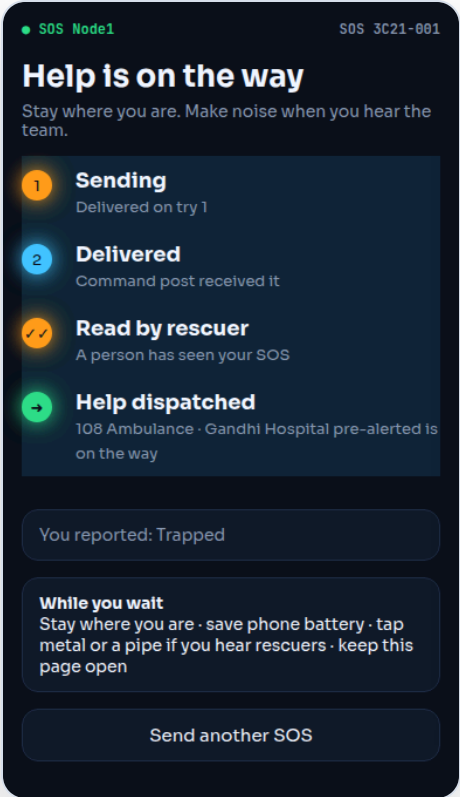


No towers · no internet · no app · no SIM · no cloud. The radio is LoRa at 433 MHz: kilometres of range on milliwatts.

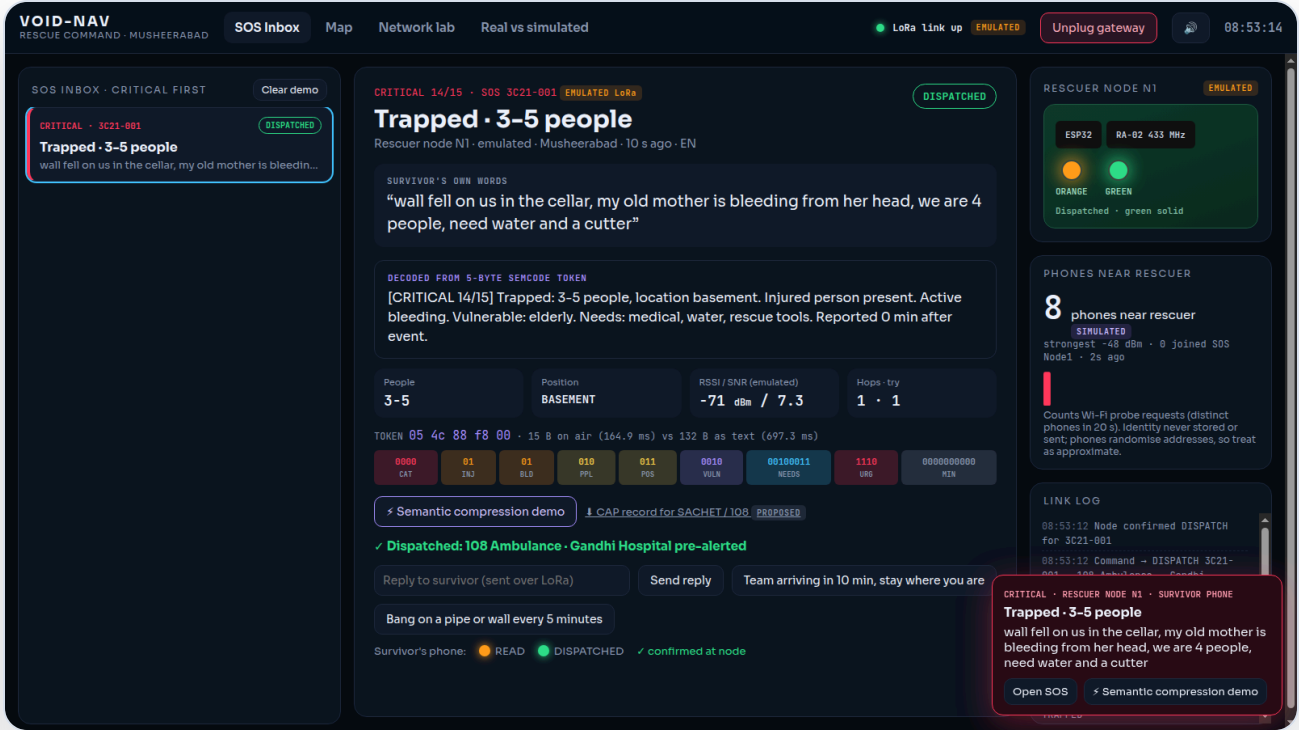
The survivor sends; the rescuer answers



Joins “SOS Node1”



Sees Read → Help on the way



Command dashboard: critical-first inbox, survivor's own words, decoded report, 108 dispatch

- 1

Delivered
Node retries up to 8 times until command ACKs.
- 2

Read
Opening the card marks it read. It cannot be undone.
- 3

Dispatched
Team or 108 ambulance plus hospital pre-alert.
- 4

Reply
“Team arriving in 10 min” pops up on the phone.
- VOID-NAV

Send the meaning, not the words

“wall fell on us in the cellar, my old mother is bleeding from her head, we are 4 people, need water and a cutter”

132 bytes typed by the survivor

↓ on-device AI reads it

Trapped

Bleeding

Basement

3-5 people

Elderly

Water · Medical · Rescue tools

Urgency 14/15

↓ SemCode codebook

05 4C 88 F8 00

Free text

142 bytes · 738 ms on air

SemCode

15 B · 165 ms

-89%

bytes

-78%

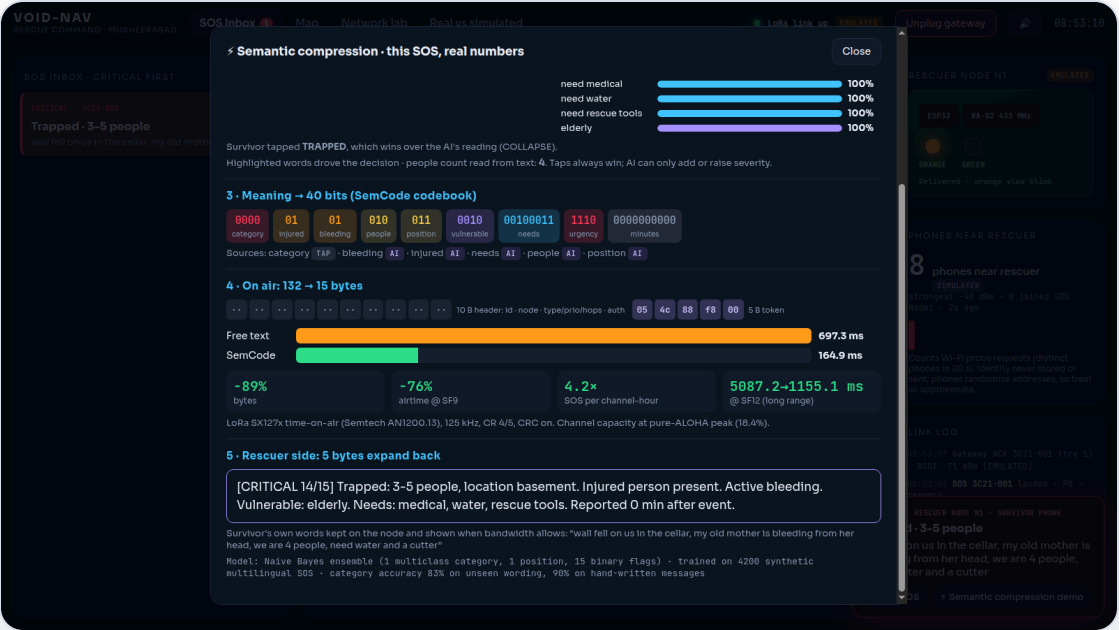
airtime

4.5×

more SOS per channel

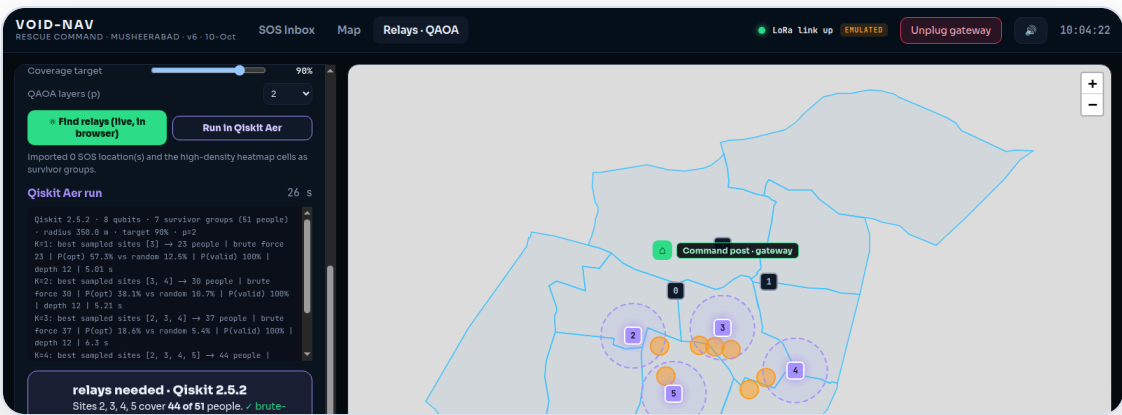
83%

category accuracy, unseen wording



Every SOS has a “Semantic compression demo” button showing its own real numbers. The model is a multilingual Naive Bayes ensemble we trained (17 heads, English/Hindi/Telugu/Hinglish). It is deterministic and never invents details; the survivor's taps always win.

Where to put the relays, and what happens when one breaks

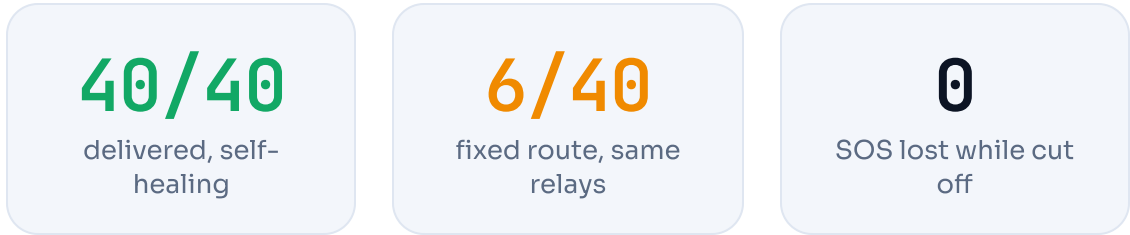


Q **QAOA answers “how many relays, and where?”**
One qubit per candidate rooftop. The circuit only explores plans with exactly K relays and favours plans that cover more people. A classical optimiser tunes it. Run in **Qiskit Aer** (8,192 shots) and checked against brute force.

K RELAYS	PEOPLE COVERED	BRUTE FORCE	QAOA FINDS BEST	RANDOM
2	35 / 64	35 ✓	36%	11%
3	50 / 64	50 ✓	16%	4%
4	59 / 64 (92%)	59 ✓	12%	1.4%



H **Self-healing: no central control**
Every node beacons “I’m N hops from the gateway”. Each hands an SOS only to a closer neighbour, which must acknowledge it. With no ACK it tries the next-best neighbour; with no route at all it stores the SOS and carries it. Nothing is dropped.



Simulated with the firmware's own rules. Team simulator: 119/119 vs 29/119.

₹1,200 nodes. Minutes after rescuers arrive. Every phone becomes a beacon that gets an answer.

Built and working

Survivor SOS page · rescuer-node firmware (Wi-Fi + LoRa + phone scan) · gateway firmware · offline command dashboard · trained AI compressor · QAOA planner in Qiskit · self-healing simulator.

Shown in simulation (labelled)

Multi-hop relay chain (we own one relay) · survivor-density heatmap of the scenario · the LoRa link when the boards are not on the table.

Proposed

CAP export to SACHET / state control rooms / 108 when any backhaul exists · encryption · GPS per node · 865 MHz modules for production.

₹1,200

per node, about 100 per satellite handset

0

internet, SIM, app or cloud needed

-78%

airtime per SOS with AI compression

40/40

SOS delivered through relay failures

Customers: NDRF · SDRF · district disaster authorities · fire services · large events · defence.